

EXPERIMENT 23: FAMILY TREE

Aim

To write a **Prolog program to implement a family tree** and define relationships such as mother, father, grandfather, grandmother, sister and brother.

Algorithm

1. Start.
2. Define male and female persons.
3. Define parent relationships.
4. Define the mother and father relations.
5. Define grandfather and grandmother.
6. Define sister and brother.
7. Execute queries to identify relationships.
8. Display the results.
9. Stop.

Source Code

```
% Family Tree  
female(pam).  
female(liz).  
female(ann).  
female(pat).  
male(tom).  
male(bob).  
male(jim).  
parent(pam, ann).  
parent(tom, ann).  
  
parent(pam, pat).
```

parent(tom, pat).

parent(ann, jim).

parent(bob, jim).

mother(X, Y) :-

female(X),

parent(X, Y).

father(X, Y) :-

male(X),

parent(X, Y).

grandfather(X, Y) :-

male(X),

parent(X, Z),

parent(Z, Y).

grandmother(X, Y) :-

female(X),

parent(X, Z),

parent(Z, Y).

sister(X, Y) :-

female(X),

parent(P, X),

parent(P, Y),

$X \neq Y$.

brother(X, Y) :-

male(X),

parent(P, X),

parent(P, Y),

X \= Y.

Queries and Output

?- mother(pam, ann).

true.

?- father(tom, pat).

true.

?- sister(ann, pat).

true.

?- brother(jim, ann).

true.

Output

```
?- [family_tree].
% family_tree.pl compiled 0.00 sec, 25 clauses

?- mother(pam, ann).
true.

?- father(tom, pat).
true.

?- sister(ann, pat).
true.

?- brother(jim, ann).
true.

?- 
```

Result

The Prolog program successfully implements the family tree and determines relationships such as **mother**, **father**, **grandfather**, **grandmother**, **sister** and **brother**.

